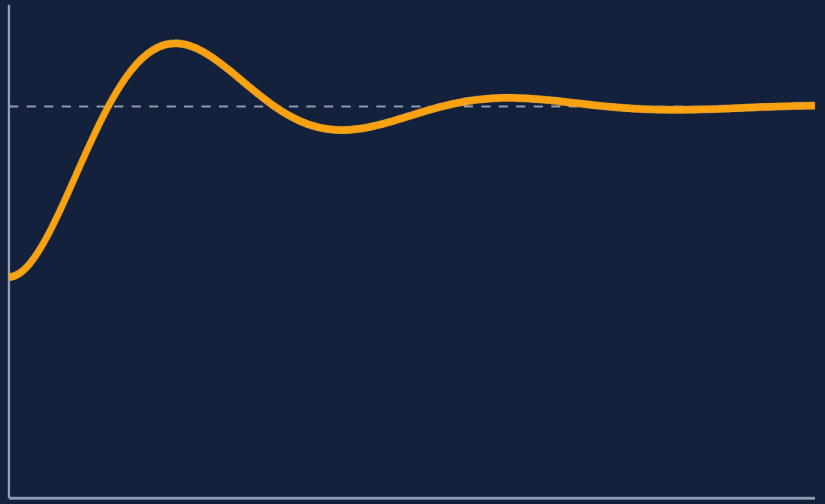
State-Variable Analysis



Control Systems (22EEG53)

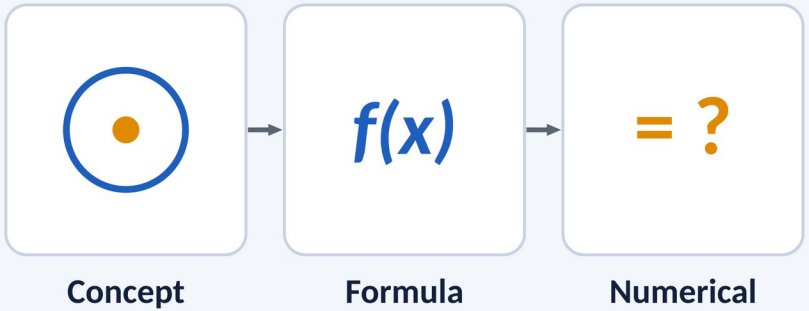
Unit 5: Compensators and Controllers

Topic: Introduction to state-variable analysis; state-space model of physical systems (SISO)

Prof. Mitesh Astik

Learning Outcomes

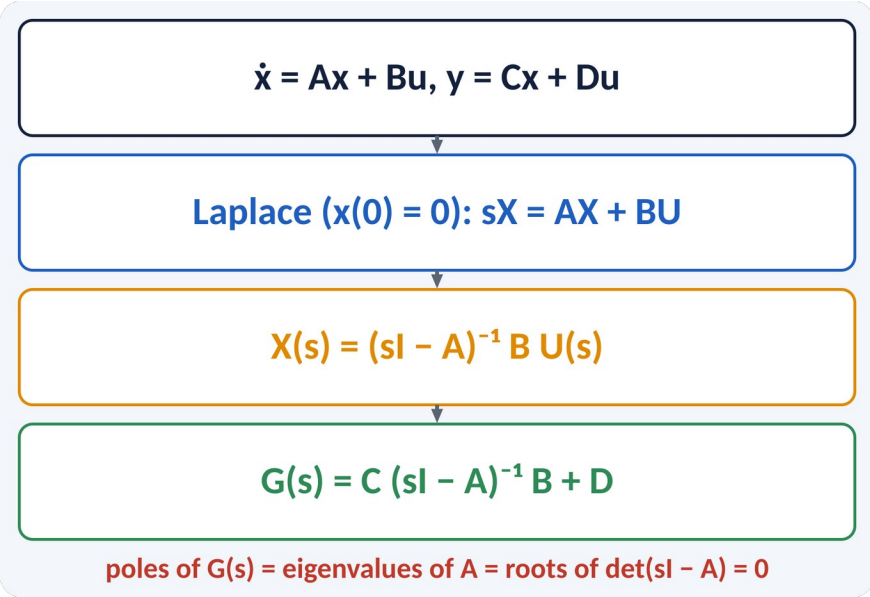
- Define state, state variable and state space.
- Write $\dot{x} = Ax + Bu$, $y = Cx + Du$.
- Model RLC and mass–spring–damper systems in state space.
- State the advantages over the transfer function.



Linked to CO5
Analyse the system using state-space techniques.

Recap: What You Already Know

- An nth-order ODE needs n initial conditions.
- Transfer function: input–output only.
- Matrix multiplication and inverse.
- Poles = roots of the characteristic equation.



Series RLC: $L\,q'' + R\,q' + q/C = v$; $M\,x'' + B\,x' + K\,x = f$

Motivation: Beyond the Transfer Function

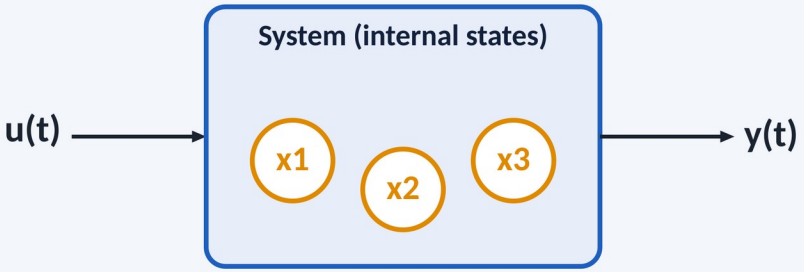
- Robots and aircraft have many inputs and outputs.
- Nonlinear and time-varying systems.
- Need to know internal quantities, e.g. current.
- Computers solve first-order matrix equations easily.

	Transfer function	State space
Systems	SISO, LTI	MIMO, nonlinear, time-varying
Initial conditions	taken as zero	included
Internal states	hidden	available
Method	s-domain algebra	matrices, computer
Design tools	Bode, root locus	modern control

State space = the language of modern control, simulation and estimation

State, State Variables, State Space

- State: the smallest set of variables describing the system.
- State variables: x1, x2, ..., xn.
- State vector x(t): the column of states.
- State space: the n-dimensional space of x.
- Number of states = order of the system.

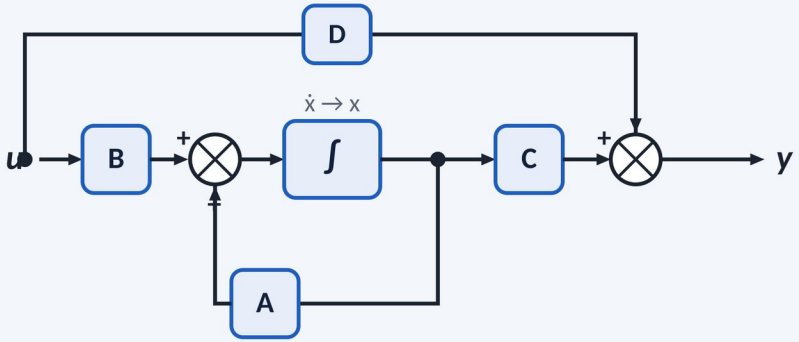


States $x(t) = [x1\ x2\ \dots\ xn]^T$: the minimum set of variables that, with the input, fixes the future behaviour

$x(t) = [x1\ x2\ \dots\ xn]^T$, n = number of energy-storage elements

State Equations

- State equation: $\dot{x} = A\,x + B\,u$.
- Output equation: $y = C\,x + D\,u$.
- A: system matrix (n × n).
- B: input matrix; C: output matrix; D: feedthrough.
- SISO: u and y are scalars.

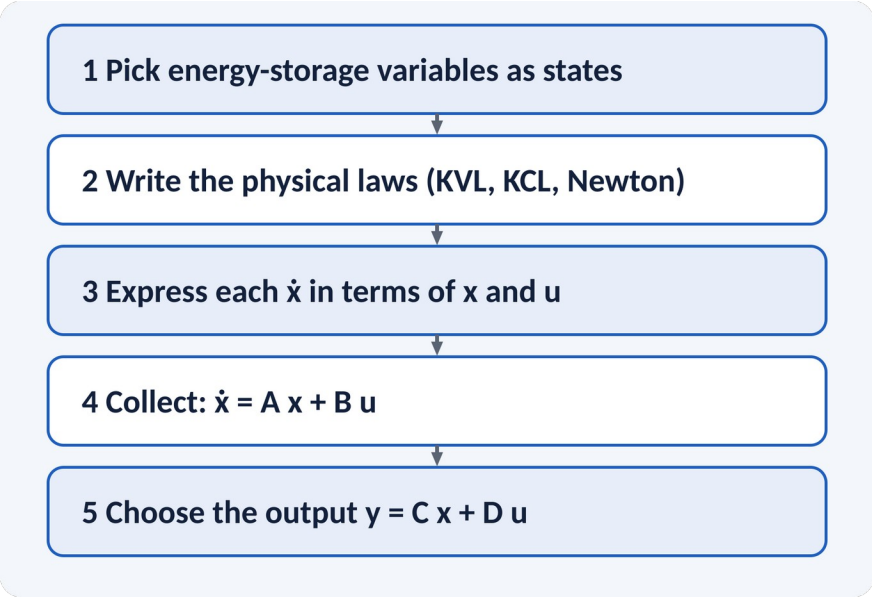


$$\dot{x} = A\,x + B\,u \quad y = C\,x + D\,u$$

$\dot{x} = A\,x + B\,u$; $y = C\,x + D\,u$
A n×n, B n×1, C 1×n, D 1×1 for a SISO system

Physical Variable Approach

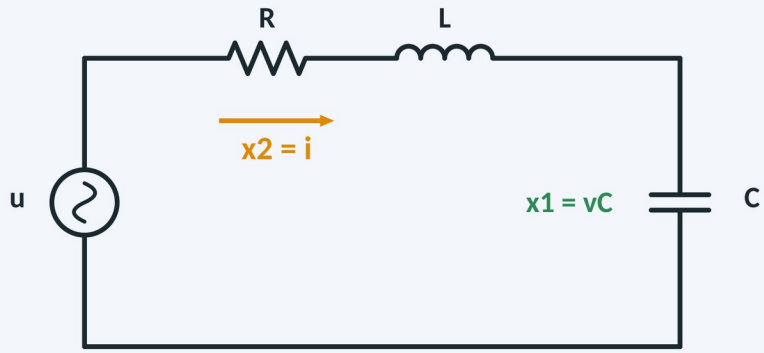
- Choose physically meaningful states.
- Energy-storage elements give the states.
- Write the governing physical laws.
- Rearrange into first-order form.
- Read off the matrices.



Electrical: x = vC, iL; mechanical: x = position, velocity

State Model of a Series RLC Circuit

- States: x1 = vC, x2 = iL = i.
- Capacitor: i = C·dvC/dt → ẋ1 = x2/C.
- KVL: u = R i + L di/dt + vC.
- Solve: ẋ2 = −x1/L − (R/L)x2 + u/L.
- Output: y = vC = x1.

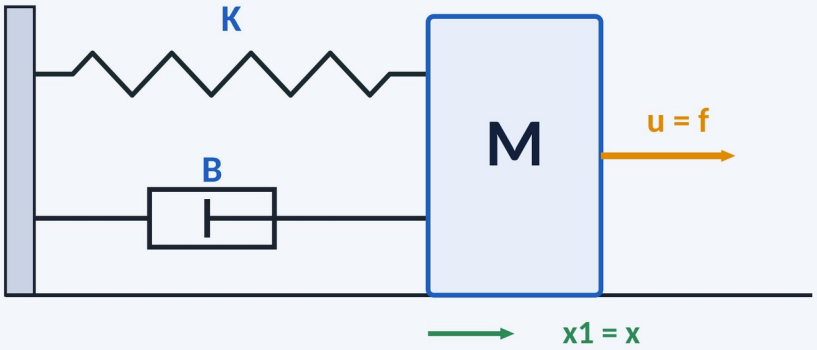


$$\dot{x}1 = x2 / C$$
$$\dot{x}2 = -x1/L - (R/L)x2 + u/L$$

$$A = [[0, 1/C], [-1/L, -R/L]], \quad B = [0, 1/L]^T, \quad C = [1 \ \ 0]$$

State Model of Mass–Spring–Damper

- States: x1 = x (position), x2 = ẋ (velocity).
- ẋ1 = x2 by definition.
- Newton: M ẍ = u − B ẋ − K x.
- Solve for ẋ2.
- Output: y = x = x1.



$$\dot{x}1 = x2$$

$$\dot{x}2 = -(K/M)x1 - (B/M)x2 + u/M \quad (x2 = \dot{x})$$

$$A = [[0, 1], [-K/M, -B/M]], \quad B = [0, 1/M]^T, \quad C = [1 \ 0]$$

Advantages of State Space

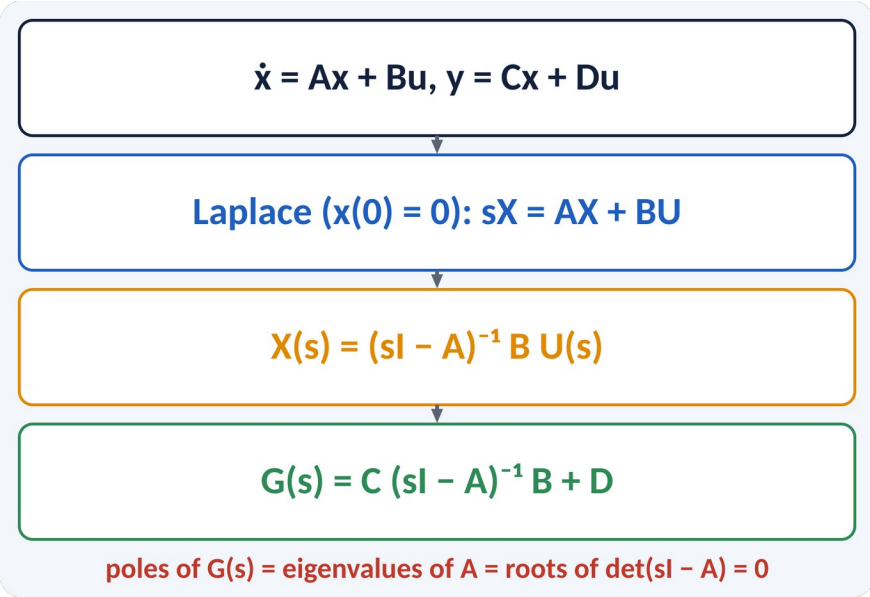
- Handles MIMO systems naturally.
- Works for nonlinear and time-varying systems.
- Includes initial conditions.
- Gives access to every internal state.
- Easy to simulate and design by computer.

	Transfer function	State space
Systems	SISO, LTI	MIMO, nonlinear, time-varying
Initial conditions	taken as zero	included
Internal states	hidden	available
Method	s-domain algebra	matrices, computer
Design tools	Bode, root locus	modern control

Transfer function: external description; state space: internal description

From State Model to Transfer Function

- Take the Laplace transform with x(0) = 0.
- Solve for X(s).
- Substitute into the output equation.
- Poles of G(s) are the eigenvalues of A.



$G(s) = C\,(sI - A)^{-1} B + D$; characteristic equation: $\det(sI - A) = 0$

Solved Example 1: RLC State Model

- R = 2 Ω, L = 1 H, C = 0.25 F; output vC. Find A, B, C and G(s).
- 1. x1 = vC, x2 = i: $\dot{x}1 = 4x2$, $\dot{x}2 = -x1 - 2x2 + u$.
- 2. $A = [[0, 4], [-1, -2]]$, $B = [0, 1]^T$, $C = [1\ 0]$, $D = 0$.
- 3. $sI - A = [[s, -4], [1, s + 2]]$, $\det = s^2 + 2s + 4$.
- 4. $(sI - A)^{-1}B = [4, s]^T/\det$.
- 5. $G(s) = C(sI - A)^{-1}B = 4/(s^2 + 2s + 4)$.

$$A = \begin{bmatrix} 0 & 4 \\ -1 & -2 \end{bmatrix} \qquad D = 0$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$\det(sI - A) = s^2 + 2s + 4$

$G(s) = 4/(s^2 + 2s + 4)$

eigenvalues: $-1 \pm j1.73$

x1 = vC, x2 = i

Answer: $G(s) = 4/(s^2 + 2s + 4)$; eigenvalues $-1 \pm j1.73$

Solved Example 2: Mass–Spring–Damper

- M = 2 kg, B = 4 N·s/m, K = 8 N/m; output x. Find the state model and G(s).
- 1. $x1 = x$, $x2 = \dot{x}$: $\dot{x}1 = x2$, $\dot{x}2 = -4x1 - 2x2 + 0.5u$.
- 2. $A = [[0, 1], [-4, -2]]$, $B = [0, 0.5]^T$, $C = [1\ 0]$.
- 3. $\det(sI - A) = s^2 + 2s + 4$.
- 4. $G(s) = 0.5/(s^2 + 2s + 4) = 1/(2s^2 + 4s + 8)$.
- 5. Eigenvalues $-1 \pm j1.73 \rightarrow \zeta = 0.5$, $\omega n = 2$ rad/s.

$$A = \begin{bmatrix} 0 & 1 \\ -4 & -2 \end{bmatrix} \qquad D = 0$$

$$B = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$\det(sI - A) = s^2 + 2s + 4$

$G(s) = 0.5/(s^2 + 2s + 4)$

eigenvalues: $-1 \pm j1.73$

$x1 = x$, $x2 = \dot{x}$ (M = 2, B = 4, K = 8)

Answer: $G(s) = 1/(2s^2 + 4s + 8)$; same poles as the RLC system (analogy)

Common Mistakes and Exam Tips

Common mistakes

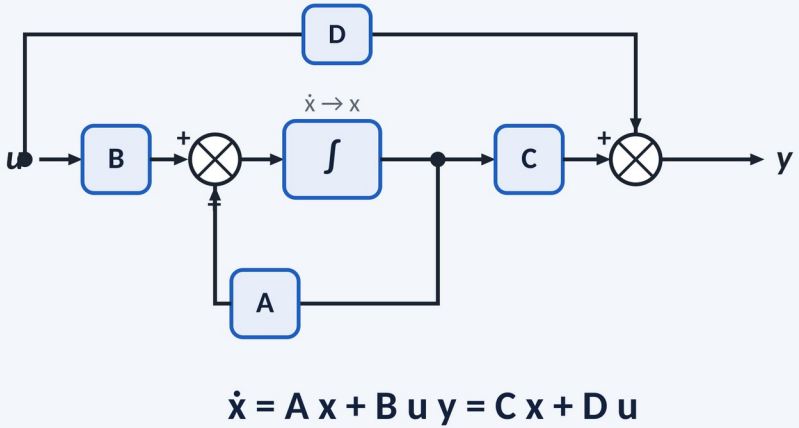
- Choosing too many or too few states.
- Mixing up the order of rows in A.
- Using initial conditions when finding G(s).
- Forgetting the 1/M factor in B.
- Confusing A (system matrix) with the open-loop transfer function.

Exam tips

- Count the energy-storage elements.
- Write $\dot{x}1 = x2$ first, then the second law.
- Check G(s) against the transfer function method.
- Check eigenvalues of A = poles of G(s).

Summary

- State vector x = minimum set describing the system.
- $\dot{x} = Ax + Bu$; $y = Cx + Du$.
- Physical approach: use energy-storage variables.
- $G(s) = C(sI - A)^{-1}B + D$; poles = eigenvalues of A.
- State space handles MIMO and nonlinear systems.



Number of states = order of the system = number of storage elements

Quick Quiz

Q1. Number of state variables of a series RLC circuit:
(a) 1 (b) 2 (c) 3 (d) 4

Ans: (b)

Q2. In $\dot{x} = Ax + Bu$, matrix A is:
(a) input matrix (b) output matrix (c) system matrix (d) feedthrough

Ans: (c)

Q3. Poles of G(s) equal:
(a) eigenvalues of A (b) elements of B (c) rows of C (d) D

Ans: (a)

Q4. States chosen for a mass–spring–damper:
(a) force and mass (b) position and velocity (c) K and B (d) energy only

Ans: (b)

Q5. A main advantage of state space:
(a) only SISO (b) handles MIMO and initial conditions (c) needs no matrices (d) only linear

Ans: (b)

References

1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **J. J. D'Azzo and C. H. Houpis, Feedback Control System Analysis and Synthesis**
   4th Ed., McGraw Hill International, 1999 (reference, Unit 5)

3. **M. Gopal, Control Systems Principles and Design**
   4th Ed., TMH, 1999 (reference, Units 4, 5)

Next lecture: course revision